Curriculum Vitae – HUGO NOWACKI

WORK EXPERIENCE

- Astrophysics Postdoc **2 years (03/2025 – 12/2026)**
OCA, Nice, France.
Principal Investigator of the « Exoplanet Host Star » science program of the Interferometric Survey of Stellar Parameters (ISSP) project.
Responsible for the characterization and optimization of the SPICA-FT fringe tracker at the CHARA array observatory.
Supervision: Denis Mourard.

- Astrophysics Ph.D. **3 years (12/2021 – 12/2024)**
IPAG, Grenoble, France
"Contribution of the VLTI/GRAVITY+ instrument to the study of interactions between young stars and their proto-planetary disk".
Supervision: Karine Perraut, Jean-Baptiste Le Bouquin.

- System engineer **1 year (11/2020 – 11/2021)**
IPAG, Grenoble, France
System engineer responsible for the low frequency radar JuRa onboard the Juventas nanosatellite for the ESA/Hera mission.
Supervision: Alain Hérique, Yves Rogez.

- Astrophysics internship **4 months (03/2020 – 07/2020)**
IPAG, Grenoble, France
"Mapping the magnetic field of the T Tauri star S CrA N through spectropolarimetric analysis".
Supervision: Evelyne Alecian, Karine Perraut.

- Thermal control engineering internship **6 months (03/2019 – 09/2019)**
IAS, Orsay, France
Designing the thermal control system for the hyperspectral imager MacrOmega onboard the JAXA/MMX mission.
Supervision: Vincent Hamm.

- Solar physics internship **3 months (05/2018 – 08/2018)**
University of Sydney, Australia
Magneto-hydro-dynamic models of solar active regions and observational analysis of flare-related photospheric phenomena.
Supervision: Michael Wheatland.

EDUCATION

- Ph.D. in astrophysics, Université Grenoble Alpes, France **2021 – 2024**

- Master's in astrophysics, Université Grenoble Alpes, France **2019 – 2020**

- Master's in astronomical and space-based system engineering (OSAE), observatoire de Meudon, France **2018 – 2019**

- Bachelor's in fundamental physics, Université Paris-Sud, France **2016 – 2018**

CONTRIBUTIONS TO CONFERENCES AND WORKSHOPS

• Invited presentations

- Annual meeting of the AG (~100 people) **September 2025, Görlitz, Germany**
Invited splinter talk: “Star-disk interactions in young Sun-like stars probed with VLTI/GRAVITY” for the Astronomische Gesellschaft.

- Observatory of Geneva seminar (~30 people) **May 2024, Geneva, Switzerland**
Invited seminar: “The innermost regions of young Sun-like stars”

- PROMETHEE ANR workshop (~20 people) **November 2023, Lyon, France**
Invited talk: “Spectropolarimetric analysis of S CrA N and constraints from interferometry for the study of protostars”

• Contributed presentations

- SHARPS Science meeting (~50 people) **July 2026, Exeter, England**
Contributed talk: “Interferometric observation of exoplanet host stars”.

- CHARA science meeting (~50 people) **Mars 2026, Socorro, New Mexico**
Contributed talk: “Interferometric observation of exoplanet host stars”.

- Cool Stars 22 (>500 people) **June 2024, San Diego, California**
Contributed plenary talk: “Perspectives on magnetospheric accretion in Classical T Tauri Stars”.

- GRAVITY+, impact on star and planet formation (50 people) **June 2024, Grenoble, France**
Member of the LOC + contributed talk: “Towards real-time magnetospheric accretion monitoring with VLTI/GRAVITY+”

- GRAVITY consortium workshop (~20 people) **November 2023, Heidelberg, Germany**
Contributed talk: “Is S CrA N growing its radiative core ? Contribution of spectropolarimetry to the study of YSOs”

• Posters

- The inner disk of young stars (~100 people) **May 2023, Cargèse, France**
Poster: “Inner regions of the strongly accreting T Tauri Star S CrA N”

- SPIE telescopes + instrumentation (>500 people) **July 2022, Montreal, Canada**
Poster: “Impact of local turbulence on high-order adaptive optics”

- Cool Stars 21 (>500 people) **July 2022, Toulouse, France**
Poster: “Structure of the inner regions in the T Tauri Star S CrA N”

INSTRUMENTATION & DATA MANAGEMENT

- GRAVITY+ adaptive optics (GPAO) **VLTI, Chile**

Responsible for the characterization of the GPAO’s deformable mirrors and their mount.
Led their qualification in Paranal and contributed to their installation.

- SPICA-FT Fringe tracker **CHARA, California**
Responsible for the characterization and optimization of the SPICA-FT fringe tracker.
Making phase-delay tracking routinely available for observations with the CHARA Array 6-telescopes beam-combiners.

- SPICA pipeline **CHARA, California**
Contributed to the data reduction pipeline of CHARA/SPICA: improved the computation of uncertainties in the calibration process of interferometric data.

OBSERVING EXPERIENCE

• Observing proposals :

- PI of the « Exoplanet host stars » of the ISSP survey **CHARA, California**
Part of the GTO Interferometric Survey of Stellar Parameters (ISSP) obtained for the development and exploitation of SPICA.

- PI of the « First multi-chromatic imaging of young stellar objects » **CHARA, California**
Four full nights allocated for 2025B and 2026A, aiming at combining R, H and K band interferometric observations from the SPICA, MIRC-X and MYSTIC instruments (program ID=SP2)

- CoI of the SPICA GTO, PI of the « Exoplanet host stars » program **CHARA, California**
25 nights allocated per year (2021-2026) to combine SPICA, MIRC-X and MYSTIC.
The exoplanet host stars (S01) program aims at measuring the size of stars to constraint the density and composition of their surrounding planets.

- CoI of 3 proposals focusing on the magnetic field of class I protostars **CFHT, Hawaii**
28.3 hours in total allocated with the SPIRou instrument (infrared spectropolarimeter) to study the magnetic field and derive a magnetic map of class I protostars (IDs=24BF012, 25AF027, 25BF007).

• Observing nights :

- SPICA, MIRC-X and MYSTIC interferometers **CHARA, California**
2 weeks on-site (October 2025),
~600 hours of remote observations for the ISSP project in 2025/2026
Operation of the three instruments simultaneously each time.

- GRAVITY interferometer **VLTI, Chile**
1-week on-site as visitor (February 2024)
~20 remote nights for the GRAVITY YSO group from 2022 to 2024.

- Neo-Narval spectropolarimeter **TBL, France**
1-week service observations (February 2023).

- SOPHIE spectrometer **OHP, France**
1-week observations for a master’s project (December 2021).

OTHERS

TEACHINGS:

- Three different teachings over two years for Université Grenoble Alpes: Real analysis seminar (72h), thermodynamics and mechanics practical classes (24h each).

PEER-REVIEWING :

- Reviewer for the Monthly Notice of the Royal Astronomical Society.

PUBLIC OUTREACH:

- Supervised about 10 astronomy-night events, combining popularization of astronomy and sky observations for all audiences (~50 people each).
- Realized 2 podcasts explaining my Ph.D. to the general public (in French)

MENTORING ACTIVITIES:

- Supervision of an undergraduate intern (3 months internship).
- Co-supervision of an undergraduate intern (3 months), and four high school interns (1 week).
- Coach and player for a local handball team for 15 years